

PATENT LANDSCAPE REPORT: LOW ACRYLONITRILE NBR-RELATED SYNTHESIS, MODIFICATION, AND PROCESSING

1. ABSTRACT

This report synthesizes a seven-patent landscape relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related NBR processing/modification technologies. The target is defined as NBR compositions with reduced acrylonitrile content, typically sought to tune flexibility, glass transition temperature, processability, and downstream performance in tires, seals, belts, charging members, and adhesive systems. The patent set is heterogeneous: only one patent directly addresses NBR molecular-weight reduction by olefin metathesis, while the others contribute enabling chemistry or application contexts that rely on NBR as a base elastomer or functional component.

The most directly relevant disclosure is US9346908B1, which treats NBR (explicitly 38 mol % acrylonitrile in the examples) with a loaded ruthenium olefin-metathesis catalyst and 1-hexene co-olefin in chlorobenzene at room temperature for two hours to reduce molecular weight and improve processability. The catalyst is immobilized on a magnetic carrier via a grafted pyridine-containing polymer, enabling magnetic recovery and reuse. This patent is important because it demonstrates a practical route for modifying NBR properties without changing monomer feed composition, but it does not itself teach low-acrylonitrile polymerization.

The remaining patents are application- and process-oriented. US8053512B2 and EP2057205B1 focus on elastomer backbone modification and metathesis-based segment interchange, respectively, using unsaturated polymers and catalysts to alter morphology, hysteresis, or block architecture. US10317811B2 and US9740133B2 disclose charging members whose electroconductive elastic layers contain NBR and/or NBR-based particle aggregates, with explicit control of surface roughness, protrusion height, particle size, and vulcanization/polishing conditions. US11131058B2 and EP2522704B1 use carboxy-modified NBR or NBR-modified epoxy resin in adhesive/treatment systems, showing how NBR chemistry is leveraged in cord treatment and structural adhesive sheets.

Across the landscape, the dominant trends are controlled catalyst-mediated modification, solvent-based processing, magnetic catalyst recovery, and formulation engineering around NBR content rather than direct low-ACN copolymerization. For low-acrylonitrile NBR synthesis specifically, the evidence supports monomer-ratio control during copolymerization as the primary

route, while the patents here mainly provide downstream processing and modification strategies that can be combined with low-ACN base polymers.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8053512B2
- Patent Title: Sulfide modified elastomeric polymers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses backbone modification of uncrosslinked elastomeric polymers using sulfide modifiers rather than direct NBR copolymerization. The elastomer backbone is reacted with a sulfide modifier under conditions intended to improve filler interaction and reduce hysteresis in vulcanized compositions.

Relevant Experimental Evidence - No numbered examples were detected in the extracted evidence. - The disclosure emphasizes sulfide backbone modifiers for elastomeric polymers. - The patent is framed around tire-performance improvement rather than NBR synthesis.

Technical Relevance Relevant as a post-polymerization modification strategy for unsaturated elastomers, including NBR-like backbones. It does not define low-acrylonitrile NBR synthesis, but it informs how unsaturation-rich elastomers can be functionalized to tune performance.

Patent 2

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent teaches metathesis-driven segment interchange between two or more ethylenically unsaturated polymers to form meta-block copolymers. The examples use Grubbs II catalyst in hot solvent solutions, followed by precipitation and extraction to isolate the metathesized products. - Polymer dissolution temperature: 95 °C — Example 1 - Catalyst amount: 8 mg Grubbs II metathesis catalyst — Example 1 - Reaction time: 1 hour — Example 1 - Precipitation solvent: methanol — Example 1 - Polymer dissolution temperature: 105 °C — Example 2 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 2 - Reaction time: 1 hour — Example 2 - Extraction solvent: 30 ml tetrahydrofuran — Example 2 - Polymer dissolution temperature: 105 °C — Example 3 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 3 - Reaction time: 1 hour — Example 3 - Polymer dissolution temperature: 105 °C — Example 4 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 4 - Reaction time: 1 hour — Example 4 - Polymer dissolution temperature: 105 °C — Example 5 - Catalyst amount: 9 milligrams Grubbs II catalyst — Example 5 - Reaction time: 1 hour — Example 5 - Extraction solvent: 30 ml tetrahydrofuran — Example 5 - Polymer dissolution temperature: 105 °C — Example 6 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 6 - Reaction time: 1 hour — Example 6 - Polymer dissolution temperature: 105 °C — Example 7 - Catalyst loading: 8,000 ppm tungsten hexachloride catalyst — Example 7 - Co-catalyst loading: 13,000 ppm tri-n-butylmethyltin co-catalyst — Example 7 - Reaction time: 1 hour — Example 7 - Polymer dissolution temperature: 105 °C — Example 8 - Catalyst amount: 9 milligrams Grubbs II catalyst — Example 8 - Reaction time: 1 hour — Example 8 - Extraction solvent: 30 ml tetrahydrofuran — Example 8 - Polymer dissolution temperature: 105 °C — Example 9 - Catalyst loading: 8,000 ppm tungsten hexachloride catalyst — Example 9 - Co-catalyst loading: 13,000 ppm tri-n-butylmethyltin co-catalyst — Example 9 - Reaction time: 1 hour — Example 9 - Polymer dissolution temperature: 105 °C — Example 10 - Catalyst amount: 9 milligrams Grubbs II catalyst — Example 10 - Reaction time: 1 hour — Example 10 - Extraction solvent: 50 ml tetrahydrofuran — Example 10

Relevant Experimental Evidence - Example 1: HPBD and partially hydrogenated NBR were metathesized to form a meta-block copolymer, confirmed by SAXS microphase separation at 300 °C. - Example 2: Unsaturated polyurethane and poly(ethylene-co-butadiene) formed a meta-block copolymer after Grubbs II treatment, verified by FTIR. - Example 7: Poly(ethylene-co-butadiene) and poly(ethylene-co-octene-co-butadiene) were converted using tungsten hexachloride/tri-n-butylmethyltin, showing altered CRYSTAF behavior. - Example 9: SBS and poly(ethylene-co-butadiene) were metathesized to yield an ordered microphase morphology consistent with a meta-block copolymer. - Example 10: A terpolymer-like meta-block product was formed from fumaryl-modified polycarbonate, SBS, and poly(ethylene-co-butadiene), supported by NMR, TEM, and SAXS.

Technical Relevance Relevant because it demonstrates catalyst-mediated modification of unsaturated polymers, including partially hydrogenated NBR, to alter morphology and properties. While not a low-ACN synthesis route, it is useful for understanding post-polymerization transformation of NBR-containing materials.

Patent 3

- Patent Number: US9346908B1
- Patent Title: Olefin-metathesis catalysts and method for preparing low-molecular-weight ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an olefin-metathesis catalyst immobilized on a magnetic carrier and a method for reducing the molecular weight of NBR by metathesis in solution. In the examples, NBR solution is treated with 1-hexene and the loaded catalyst at room temperature, then the catalyst is magnetically separated and reused. - Acrylonitrile content: 38 mol % — Comparative Example 1 - Solid content: 6 wt % — Comparative Example 1 - Co-olefin amount: 2 wt % relative to solid content — Comparative Example 1 - Reaction time: two hours — Comparative Example 1 - Reaction temperature: room temperature — Comparative Example 1 - Drying temperature: 45° C. — Comparative Example 1 - Catalyst reuse cycles: five times — Comparative Example 1 - Molecular weight before reaction: 126,049 — Comparative Example 1 - Molecular weight after round 1: 66,190 — Comparative Example 1 - PDI before reaction: 1.98 — Comparative Example 1 - PDI after round 1: 1.81 — Comparative Example 1 - Acrylonitrile content: 38 mol % — Example 7 - Solid content: 6 wt % — Example 7 - 1-hexene amount: 2 wt % relative to solid content — Example 7 - Loaded-type catalyst amount: 5 wt % relative to solid content — Example 7 - Reaction time: two hours — Example 7 - Reaction temperature: room temperature — Example 7 - Drying temperature: 45° C. — Example 7 - Catalyst reuse cycles: five times — Example 7 - Molecular weight before reaction: 126,049 — Example 7 - Molecular weight after round 1: 45,925 — Example 7 - PDI before reaction: 1.98 — Example 7 - PDI after round 1: 1.61 — Example 7 - Acrylonitrile content: 38 mol % — Example 8 - Solid content: 6 wt % — Example 8 - 1-hexene amount: 2 wt % relative to solid content — Example 8 - Loaded-type catalyst amount: 5 wt % relative to solid content — Example 8 - Reaction time: two hours — Example 8 - Reaction temperature: room temperature — Example 8 - Drying temperature: 45° C. — Example 8 - Catalyst reuse cycles: five times — Example 8 - Molecular weight before reaction: 126,049 — Example 8 - Molecular weight after round 1: 41,583 — Example 8 - PDI before reaction: 1.98 — Example 8 - PDI after round 1: 1.59 — Example 8 - Acrylonitrile content: 38 mol % — Example 9 - Solid content: 6 wt % — Example 9 - 1-hexene amount: 2 wt % relative to solid content — Example 9 - Loaded-type catalyst amount: 5 wt % relative to solid content — Example 9 - Reaction time: two hours — Example 9 - Reaction temperature: room temperature — Example 9 - Drying temperature: 45° C. — Example 9 - Catalyst reuse cycles: five times — Example 9 - Molecular weight before reaction: 126,049 — Example 9 - Molecular weight after round 1: 39,054 — Example 9 - PDI before reaction: 1.98 — Example 9 - PDI after round 1: 1.61 — Example 9

Relevant Experimental Evidence - Example 1: Demonstrated synthesis of a pyridine-containing graft polymer used as the catalyst support layer. - Example 4: Showed loading of Grubbs

second-generation catalyst onto Fe-StVP1, giving 2.68 wt % catalyst loading. - Example 7: Reduced NBR molecular weight from 126,049 to 45,925 with magnetic catalyst recovery over five cycles. - Example 8: Achieved a stronger molecular-weight reduction to 41,583 using Fe-StVP2-GII. - Example 9: Achieved the highest first-round degradation among the catalyst-recovery tests, reducing Mw to 39,054.

Technical Relevance This is the most directly relevant patent because it explicitly processes NBR and reports its acrylonitrile content at 38 mol %. It supports a metathesis-based route to lower NBR molecular weight and improve processability, with practical catalyst recovery, but it does not disclose low-acrylonitrile polymerization itself.

Patent 4

- Patent Number: US10317811B2
- Patent Title: Charging member, method for producing same, process cartridge and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a charging member with an electroconductive elastic layer based on a butadiene-skeleton polymer, including NBR in the examples. The process involves compounding NBR with fillers and additives, extrusion molding onto a conductive support, vulcanization, and surface polishing/UV treatment to create controlled roughness and charging performance. - NBR grade: JSR N230SV — Example 1 - Amount of bonded acrylonitrile: 35.0% — Example 1 - Mooney viscosity (ML1+4 100° C.): 32 — Example 1 - Specific gravity: 0.98 — Example 1 - Carbon black amount: 48 parts by mass — Example 1 - Carbon black arithmetic mean particle diameter: 28 nm — Example 1 - Carbon black nitrogen adsorption specific surface area: 77 m²/g — Example 1 - Carbon black DBP absorbed amount: 87 cm³/100 g — Example 1 - Electro-conductive support diameter: 6 mm — Example 1 - Electro-conductive support length: 252 mm — Example 1 - Kneading temperature: 50° C. — Example 1 - Kneading time: 15 minutes — Example 1 - Second kneading temperature: 25° C. — Example 1 - Crosshead extrusion cylinder diameter: 70 mm — Example 1 - Layer thickness: 1.5 mm — Example 1 - Outer diameter of unvulcanized rubber roller: 9.0 mm — Example 1 - Vulcanization temperature: 170° C. — Example 1 - Vulcanization time: 1 hour — Example 1 - Polishing wheel grain size: 100 mesh — Example 1 - Roller rotation speed: 350 rpm — Example 1 - Ultraviolet wavelength: 254 nm — Example 1 - Integrated UV dose: 9000 mJ/cm² — Example 1 - NBR grade: JSR N230SL — Comparative Example 21 - NBR grade: JSR N215SL — Comparative Example 21 - NBR grade: JSR SL552 — Comparative Example 21 - NBR grade: JSR BR51 — Comparative Example 21 - Carbon black amount: 50 parts by mass — Comparative Example 21 - Zinc oxide amount: 5 parts by mass — Comparative Example 21 - Zinc stearate amount: 1 parts by mass — Comparative Example 21 - Calcium carbonate amount: 30 parts by mass —

Comparative Example 21 - Sulfur amount: 1 parts by mass — Comparative Example 21 - TBzTD amount: 4 parts by mass — Comparative Example 21 - Vulcanization temperature: 160° C. — Comparative Example 21 - Vulcanization time: 1 hr — Comparative Example 21 - UV integrated amount: 9000 mJ/cm² — Comparative Example 21 - UV integrated amount: 3000 mJ/cm² — Comparative Example 21 - Grinding stone grain size: 100 mesh — Comparative Example 6 - Roller rotations: 400 rpm — Comparative Example 6 - Grinding stone rotations: 2500 rpm — Comparative Example 6 - Incision rate: 20 mm/min — Comparative Example 6 - Spark out time: 1 second — Comparative Example 6 - Grinding margin: 400 µm — Comparative Example 6 - Outer diameter difference: 200 µm — Comparative Example 6

Relevant Experimental Evidence - Example 1: Demonstrated a charging roller made from NBR with 35.0% bonded acrylonitrile, vulcanized and UV-treated to obtain a roughened surface. - Comparative Example 21: Showed multiple NBR grades and formulation variations in charging-member performance testing. - Comparative Example 6: Demonstrated that grinding conditions and UV treatment can be varied to alter surface properties.

Technical Relevance Relevant because it explicitly uses NBR as the elastomeric base and provides formulation/process conditions that influence surface properties and durability. Although not a synthesis patent, it is useful for understanding how NBR composition and processing affect downstream performance in electroconductive applications.

Patent 5

- Patent Number: US9740133B2
- Patent Title: Charging member, process cartridge and electrophotographic image forming ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a charging member whose electroconductive elastic layer contains aggregates of crosslinked rubber particles, with NBR used as the elastic-layer base in many examples. The member is produced by compounding NBR with crosslinked rubber particle aggregates, molding, polishing to create protrusions, and evaluating contamination resistance and image quality. - Aggregate content: 10% by mass or more and 50% by mass or less — Claim 1 - Circle-equivalent diameter: 10 µm or more and 50 µm or less — Claim 1 - Average protrusion height: 50 nm or more and 200 nm or less — Claim 1 - Primary particle diameter: 100 nm or more and 1000 nm or less — Claim 3 - Average degree of unevenness: 1.10 or more and 1.30 or less — Claim 5 - NBR grade: JSR N230SV — Example 1 - Amount of bonded acrylonitrile: 35.0% — Example 1 - Mooney viscosity (ML1+4 100° C.): 32 — Example 1 - Specific gravity: 0.98 — Example 1 - Carbon black amount: 48 parts by mass — Example 1 - Carbon black arithmetic mean particle diameter: 28 nm — Example 1 -

Carbon black nitrogen adsorption specific surface area: 77 m²/g — Example 1 - Carbon black DBP absorbed amount: 87 cm³/100 g — Example 1 - Electro-conductive support diameter: 6 mm — Example 1 - Electro-conductive support length: 252 mm — Example 1 - Kneading temperature: 50° C. — Example 1 - Kneading time: 15 minutes — Example 1 - Second kneading temperature: 25° C. — Example 1 - Crosshead extrusion cylinder diameter: 70 mm — Example 1 - Layer thickness: 1.5 mm — Example 1 - Outer diameter of unvulcanized rubber roller: 9.0 mm — Example 1 - Vulcanization temperature: 170° C. — Example 1 - Vulcanization time: 1 hour — Example 1 - Polishing wheel grain size: 100 mesh — Example 1 - Roller rotation speed: 350 rpm — Example 1 - UV wavelength: 254 nm — Example 1 - UV integrated dose: 9000 mJ/cm² — Example 1 - Particle No.: 29 — Comparative Example 1 - Particle No.: 30 — Comparative Example 1 - Particle No.: 31 — Comparative Example 1 - Particle No.: 1 — Comparative Example 1 - Particle No.: 32 — Comparative Example 1 - Primary particle diameter: 500 nm — Comparative Example 1 - Primary particle diameter: 400 nm — Comparative Example 1 - Primary particle diameter: 400 nm — Comparative Example 1 - Primary particle diameter: 500 nm — Comparative Example 1 - Primary particle diameter: 500 nm — Comparative Example 1 - Circle-equivalent diameter: 25 μm — Comparative Example 1 - Circle-equivalent diameter: 60 μm — Comparative Example 1 - Circle-equivalent diameter: 8 μm — Comparative Example 1 - Circle-equivalent diameter: 25 μm — Comparative Example 1 - Average height of protrusion: 100 nm — Comparative Example 1 - Average height of protrusion: 250 nm — Comparative Example 1 - Average height of protrusion: 50 nm — Comparative Example 1 - Average height of protrusion: 250 nm — Comparative Example 1 - Average height of protrusion: 30 nm — Comparative Example 1 - Content ratio in electro-conductive elastic layer: 30% by mass — Comparative Example 1 - Content ratio in electro-conductive elastic layer: 8% by mass — Comparative Example 1 - Content ratio in electro-conductive elastic layer: 51% by mass — Comparative Example 1 - Number of revolutions of polishing work: 350 rpm — Comparative Example 1 - Spark-out time: 0.0 sec — Comparative Example 1 - Spark-out time: 2.5 sec — Comparative Example 1 - Spark-out time: 5.0 sec — Comparative Example 1 - Spark-out time: 10.0 sec — Comparative Example 1 - UV integrated amount: 9000 mJ/cm² — Comparative Example 1 - UV integrated amount: 3000 mJ/cm² — Comparative Example 1

Relevant Experimental Evidence - Example 1: Demonstrated a charging roller using NBR with 35.0% bonded acrylonitrile and a roughened surface formed by polishing after vulcanization. - Comparative Example 1: Showed that omitting the aggregate of crosslinked rubber particles changes the charging-member structure and performance. - Comparative Examples 2 to 8: Explored how particle type, protrusion height, and polishing conditions affect contamination resistance and image quality.

Technical Relevance Relevant because it uses NBR as the base elastomer and directly links NBR-based surface engineering to contamination suppression. It is not a synthesis route for low-ACN NBR, but it provides formulation and surface-structure guidance for NBR-containing products.

Patent 6

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses treatment agents for power-transmission belt cords based on resorcin-formaldehyde condensate, carboxy-modified latex, and polycarbodiimide resin in water. The NBR relevance is through the use of carboxy-modified acrylonitrile-butadiene copolymer latex as one of the preferred rubber components in the first treatment agent. - Proportion of condensate (A1): 10 to 100 parts by mass relative to 100 parts by mass of rubber component (A2) — Claim 1 - Polycarbodiimide resin NCN equivalent: 300 to 450 — Claim 1 - Proportion of curing agent (A3): 1 to 10 parts by mass relative to 100 parts by mass of rubber component (A2) — Claim 2 - First treatment agent proportion: 15% to 20% by mass based on entirety of first treatment agent — Detailed description - Proportion of condensate (A1): 1 to 100 parts by mass relative to 100 parts by mass of rubber component (A2) — Detailed description - Second treatment agent concentration: 5% by mass — Table 4 - RFL-X blend amount: 625 g — Table 1 - COOH-modified NBR content: 100 — Table 1 - RF condensate amount: 25 — Table 1 - Polycarbodiimide dispersion liquid B amount: 2.5 g — Table 1 - Water amount: 97.5 g — Table 1 - Total solid content concentration: 17% — Table 1 - Total solid content concentration: 18% — Table 1 - Total solid content concentration: 19% — Table 1 - Total solid content concentration: 20% — Table 1 - Total solid content concentration: 14% — Table 1 - RFL-Z blend amount: 700 g — Table 1 - Polycarbodiimide dispersion liquid A amount: 37.5 g — Table 3 - Blocked isocyanate dispersion liquid amount: 33.3 g — Table 1 - Water amount: 449.3 g — Table 1 - EPDM amount: 100 parts by mass — Table 5 - Stearic acid amount: 5 parts by mass — Table 5 - Zinc oxide amount: 5 parts by mass — Table 5 - HAF carbon black amount: 10 parts by mass — Table 5 - Hydrous silica amount: 15 parts by mass — Table 5 - RF condensate amount: 2 parts by mass — Table 5 - Anti-oxidant amount: 2 parts by mass — Table 5 - Vulcanization accelerator MBTS amount: 2 parts by mass — Table 5 - Hexamethoxy methylol melamine amount: 2 parts by mass — Table 5 - Sulfur amount: 1 part by mass — Table 5 - Toluene amount: 900 parts by mass — Table 6 - Untreated yarn immersion time: 10 seconds — Adhesion treatment

Relevant Experimental Evidence - Example 5: Showed a first treatment agent containing COOH-modified NBR and RF condensate with polycarbodiimide dispersion, including solid-content and blend ratios. - Example 9: Demonstrated alternative treatment-agent compositions using COOH-modified NBR, COOH-modified VP, and polycarbodiimide dispersion liquids. - Adhesion treatment: The untreated twisted yarn cord was immersed in the first treatment agent for 10 seconds, illustrating the cord-treatment process.

Technical Relevance Relevant because it explicitly incorporates carboxy-modified acrylonitrile-butadiene copolymer latex, showing a practical NBR-derived formulation in cord treatment. It does not address low-acrylonitrile NBR synthesis, but it is useful for downstream use of NBR-based materials in belt reinforcement.

Patent 7

- Patent Number: EP2522704B1
- Patent Title: Adhesive sheet and bonding method using same
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an adhesive sheet containing an acrylic resin, epoxy resin, and curing agent, where the epoxy resin is a mixture including NBR-modified epoxy resin. The examples prepare adhesive sheets by mixing acrylic ester copolymer resin with bisphenol A epoxy resin and NBR-modified epoxy resin, then coating and heat-curing to form laminated bodies. - Epoxy resin content: 175 to 300 parts by weight per 100 parts by weight of acrylic resin — Claim 1 - NBR-modified epoxy resin content: 25 to 75 parts by weight per 100 parts by weight of acrylic resin — Claim 6 - Acrylic resin amount: 100 parts — Example 1 - Bisphenol A epoxy resin JER828 amount: 50 parts — Example 1 - Bisphenol A epoxy resin JER1001 amount: 100 parts — Example 1 - NBR-modified epoxy resin EPR4030 amount: 50 parts — Example 1 - Dicyandiamide amount: 7 parts — Example 1 - Coating coverage: 100 g/m² — Example 1 - Adhesive sheet size: 25 mm x 12.5 mm — Example 1 - Tentative fixing load: 3 kg — Example 1 - Heat-curing temperature: 180°C — Example 1 - Heat-curing time: 1 hr — Example 1 - Acrylic resin amount: 100 parts — Example 5 - Bisphenol A epoxy resin JER828 amount: 50 parts — Example 5 - NBR modified epoxy resin EPR4030 amount: 50 parts — Example 5 - Phenoxy resin YP-50EK35 amount: 100 parts — Example 5 - Dicyandiamide amount: 7 parts — Example 5 - Adhesion strength: 20.0 MPa — Table 1 - Initial adhesive force: 15.0 N/25mm — Table 1 - Adhesion strength: 25.0 MPa — Table 1 - Initial adhesive force: 14.0 N/25mm — Table 1 - Adhesion strength: 21.4 MPa — Table 1 - Initial adhesive force: 14.0 N/25mm — Table 1 - Adhesion strength: 11.6 MPa — Table 1 - Initial adhesive force: 20.0 N/25mm — Table 1 - Adhesion strength: 15.6 MPa — Table 1 - Initial adhesive force: 14.0 N/25mm — Table 1 - Adhesion strength: 4.7 MPa — Table 1 - Initial adhesive force: 10.0 N/25mm — Table 1

Relevant Experimental Evidence - Example 1: Prepared an adhesive sheet using acrylic ester copolymer resin, bisphenol A epoxy resins, NBR-modified epoxy resin, and dicyandiamide, then heat-cured at 180°C for 1 hr. - Example 5: Added phenoxy resin to the NBR-modified epoxy formulation, showing an alternative adhesive composition. - Comparative Example 1: Replaced phenoxy resin with polyvinyl formal and measured lower adhesion performance.

Technical Relevance Relevant because it uses NBR-modified epoxy resin as a key formulation component, demonstrating a downstream application of NBR chemistry. It does not synthesize low-acrylonitrile NBR, but it shows how NBR-derived functionality is leveraged in structural adhesives.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US9346908B1 is the only patent that directly processes NBR as the target polymer; it uses a 38 mol % acrylonitrile NBR solution, 1-hexene co-olefin, and a metathesis catalyst to reduce molecular weight rather than to alter ACN content.
- US8053512B2 and EP2057205B1 both rely on unsaturated elastomer backbones and catalyst-mediated modification, but they focus on backbone functionalization or segment interchange, not monomer-ratio control during NBR synthesis.
- US10317811B2 and US9740133B2 both use NBR in electroconductive charging members; the disclosed NBR in these examples has 35.0% bonded acrylonitrile, and performance is governed by filler/particle morphology, vulcanization, polishing, and UV treatment.
- US11131058B2 and EP2522704B1 use NBR-derived materials in adhesive/treatment formulations, specifically carboxy-modified NBR latex and NBR-modified epoxy resin, emphasizing formulation compatibility rather than polymerization conditions.
- Across the set, no patent explicitly discloses low-acrylonitrile NBR copolymerization feed ratios or emulsion/solution polymerization recipes for making low-ACN NBR; the evidence is strongest for post-polymerization modification and application engineering.
- Process conditions that recur in the relevant patents include room-temperature metathesis in chlorobenzene/toluene, 95-105 °C catalyst-loading or segment-interchange steps, methanol precipitation, THF extraction, and vulcanization/heat-curing steps for NBR-containing formulations.

4. CONCLUSION

The evidence base shows that low-acrylonitrile NBR itself is not directly synthesized in the provided patents; instead, the landscape is dominated by NBR post-processing, catalyst-mediated molecular-weight reduction, and NBR-based formulation engineering. The most technically relevant disclosure is US9346908B1, which demonstrates that NBR at 38 mol % acrylonitrile can be efficiently degraded by olefin metathesis using a recoverable magnetic catalyst system, improving processability while preserving narrow PDI. For low-acrylonitrile NBR synthesis, the patents do not provide direct monomer-feed recipes, so the recommended synthesis strategy remains controlled copolymerization of butadiene with a reduced acrylonitrile feed, followed by careful control of conversion and molecular weight. Based on the broader evidence, practical downstream processing parameters include solvent-based handling, catalyst recovery where applicable, and formulation tuning via fillers, particle aggregates, and curing conditions to exploit

the softer, more flexible character expected from lower-ACN NBR grades.

5. REFERENCES

- US8053512B2 | Sulfide modified elastomeric polymers | Not disclosed | US | 2011 | <https://patents.google.com/patent/US8053512B2/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- US9346908B1 | Olefin-metathesis catalysts and method for preparing low-molecular-weight ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9346908B1/en>
- US10317811B2 | Charging member, method for producing same, process cartridge and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10317811B2/en>
- US9740133B2 | Charging member, process cartridge and electrophotographic image forming ... | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9740133B2/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>
- EP2522704B1 | Adhesive sheet and bonding method using same | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2522704B1/en>